Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/31/2012
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	A Quantitative Test of On-Orbit Exercise Countermeasures for Bone Demineralization Using a Bed Rest Analog-NNX08BA58G		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cavanagh@u.washington.edu	**Fax:**	FY 206-685-3139
PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2008	**End Date:**	05/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
Contact Email:	linda.loerch-1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2012 per NSSC information (Ed., 12/21/2011) NOTE: Change in Risk/Gaps per HRR information (Ed., 9/23/2011) NOTE: End date will be 2/29/2012 per C. Guidry/JSC (Ed., 8/9/2011) NOTE: End date is now 1/15/2013 per JSC HRP Master Task List 5/20/2011 (Ed., 7/21/2011) NOTE: Received funding augmentation/new end date to 8/31/2011 (per NSSC info 2/2010) NOTE: Received NCE to 12/31/2010 per JSC (8/09) NOTE: Received NCE to 7/31/2009 (per NSSC info; added 8/09) NOTE: Received NCE to 4/30/2009; original end date was 12/31/2008 (per NSSC info; added 8/09)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Licata, Angelo (Cleveland Clinic) Ilaslan, Hakan (Cleveland Clinic)		
Grant/Contract No.:	NNX08BA58G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Bedrest is an excellent analog for the study of on-orbit musculoskeletal changes that occur during long-duration spaceflight. Our novel approach involves quantifying typical daily loads to the lower extremities in selected free-living volunteer subjects who are moderate exercisers and planning individual exercise prescriptions based on daily load stimulus (DLS) theory. The specific aim of this project is to determine if individualized mechanical load replacement and supplementation during bedrest can serve as a countermeasure to losses in bone mineral density (BMD), bone quality, and muscle atrophy during 12 weeks of bedrest.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained as a result of this study will provide further insight into the role of skeletal loading in bone loss during long-duration spaceflight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	SUPPLEMENTAL REPORTING FOR FINAL REPORT--SEPTEMBER 2012 The total number of subjects who have completed the study at the Cleveland Clinic is 11 (6 controls and 5 exercisers). During the last year, thorough data analysis was completed. Samples were batch analyzed at the NASA Nutritional Biochemistry Lab in Houston, TX. Preparation of a manuscript for publication is underway. The results of this study support the conclusion that the individualized replacement of daily load through 1-g loaded locomotor exercise attenuates, but does not prevent, changes to the musculoskeletal system during bed rest. ANNUAL REPORTING--SEPTEMBER 2011 The total number of subjects who have completed the study at the Cleveland Clinic is 11 (6 controls and 5 exercisers). During the last year, the project team has worked to overcome administrative and regulatory procedures to allow the completion of the study at the NASA Flight Analogs Research Unit at the University of Texas Medical Branch in Galveston, TX. The study has recently been de-scoped such that no additional subjects will be enrolled. The study was on hold for several months to allow the de-scoping process to be completed and close-out procedures are now moving forward. Samples are being prepared for shipment to the NASA Nutritional Biochemistry Lab in Houston, TX for batch analysis of the bone biomarkers.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Cavanagh PR, Rice J, Novotny S, Hanson AM, Gopalakrishnan R, Genc KO, Englehaupt RK, Davis BL, Comstock B, Ilaslan H, Licata AA. "Replacing Daily Loads During 84-Days of Bedrest: An Interim Report." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Cavanagh PR, Rice AJ, Licata AA, Kuklis MM, Novotny SC, Genc KO, Englehaupt RK, Hanson AM. "A novel lunar bed rest analogue." Aviat Space Environ Med. 2013 Nov;84(11):1191-5. PubMed PMID: 24279234 ; http://dx.doi.org/10.3357/ASEM.3472.2013 , Nov-2013
Articles in Peer-reviewed Journals	Cavanagh PR, Rice AJ, Novotny SC, Genc KO, Englehaupt RK, Owings TM, Comstock B, Cardoso T, Ilaslan H, Smith SM, Licata AA. "Replacement of daily load attenuates but does not prevent changes to the musculoskeletal system during bed rest." Bone Reports. 2016 Dec;5:299-307. http://dx.doi.org/10.1016/j.bonr.2016.10.001 , Dec-2016
Dissertations and Theses	Genc KO. "The Effects of Altered Gravity Environments on the Mechanobiology of Bone: From Bedrest to Spaceflight." Dissertation, Case Western University, June 2011. , Jun-2011